

DIGITAL RADIO MONDIALE

And How To Receive It

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Digital Radio Mondiale technology uses digital signals to transmit in long-wave (190kHz - 250kHz), medium-wave (550kHz - 1600kHz) and short-wave frequencies (below 30MHz)

Digital Radio Mondiale (DRM) is a new technology in high-quality audio transmission. It is considered superior to AM and FM transmissions, as it delivers near-FM quality audio using digital audio encoding. DRM technology uses digital signals to transmit in long-wave (190kHz - 250kHz), medium-wave (550kHz - 1600kHz) and short-wave frequencies (below 30MHz). The signals provide data content such as broadcast schedules, news, weather and traffic information, among others, which could be displayed on a DRM receiver.

Apart from high-quality audio, DRM transmitters require less power compared to analogue ones. The same transmitter can be used to transmit both analogue and DRM transmissions, simultaneously. These factors make DRM a futuristic technology, to be adopted by several international broadcasters including All India Radio (AIR).

AIR has been silently experimenting with DRM transmissions since 2015. Initially, transmissions were carried out from New Delhi (Khampur) and Bengaluru, using different short-wave frequencies targeting the UK and western Europe (9950kHz), Australia and New Zealand (11,620kHz and 17,895kHz), China, Korea and Japan (11,645kHz and 13,605kHz), north east Asia (15,795kHz), Russia, Ukraine and surrounding regions (15,140kHz), Sri Lanka (15,050kHz), and Mauritius and East Africa (17,715kHz).

Having successfully completed DRM transmissions in short wave, AIR started testing these in medium wave covering

33 major cities across India. The list of the cities where DRM transmissions are currently going on can be found on AIR's website.

Countries like the UK (BBC World Service @ 15,620kHz), Kuwait (Radio Kuwait @ 11,970kHz) and Romania (Radio Romania @ 9740kHz) are also transmitting DRM signals in short wave, which can be received in India.

We need a DRM receiver to receive DRM transmissions. Although, several manufacturers are selling DRM receivers, none has started production on a large scale. DRM transmissions across the world are mostly in the testing stage, and awareness about DRM among people is abysmally low.

Currently, a DRM receiver is available in the range of ₹ 15,000 to ₹ 20,000. Prices would fall if more manufacturers enter the field, after AIR starts its regular transmission across the country, from its current stage of testing. Figs 1, 2 and 3 show some DRM receivers.

Software-defined radios (SDRs) are available at cheaper prices online, which can be used to receive DRM transmissions. Some commercial wide-band SDRs are RTL-SDR V3 (cheapest), FUNcube Pro +, Airspy, SDRPlay, HackRF and BladeRF.

SDRs are used for radio communication based on software-defined wireless communication protocol. Instead of components (such as radio frequency amplifiers and mixers), these use software on embedded or computer systems.

An SDR can be simply plugged into the USB port of a laptop or desktop computer.



Fig. 1: Gospell DRM receiver



Fig. 2: Titus II DRM receiver



Fig. 3: New Star DRM receiver